

Deploying a FAIR-Compliant Research Data Management System with iRODS at LCSB, University of Luxembourg

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University of Luxembourg | iRODS UGM 2026

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About LCSB

DATA SOURCES

Luxembourg Centre for Systems Biomedicine, University of Luxembourg

1

Data coordination centre

We coordinate data for multi-site research projects

2

8 scientific platforms

Genomics, metabolomics, bioimaging, biobank, Bioinformatics etc. — high-volume data generators

3

Research labs

Standard project files generated across individual labs

4

Clinical studies

Clinical data, including studies we support for partners

SCALE TODAY

>1 PB

managed today — mostly
imaging

12 PB

Dell EMC Isilon capacity



More than a research institute — a data coordination centre, platforms, labs and clinical studies, all on one governed platform.

The Challenge

BEFORE

Data scattered across HPC, drives & email

No single place to find, access or share

Data lost when researchers leave

Compliance manual & error-prone

WITH LCSB DATA-SPACE

our institutional platform for governed research data

One governed, structured, auditable home

Find, access & share — policy-based access

Retained & owned — never lost

Compliance & governance built in

From scattered files to one governed, auditable home for research data.

Why iRODS?



Policy-Based Management

Rule engine enforces logic at the data layer — not inside every application.



Backend Agnostic

Dell EMC Isilon 12 PB today; storage is pluggable — no vendor lock-in.

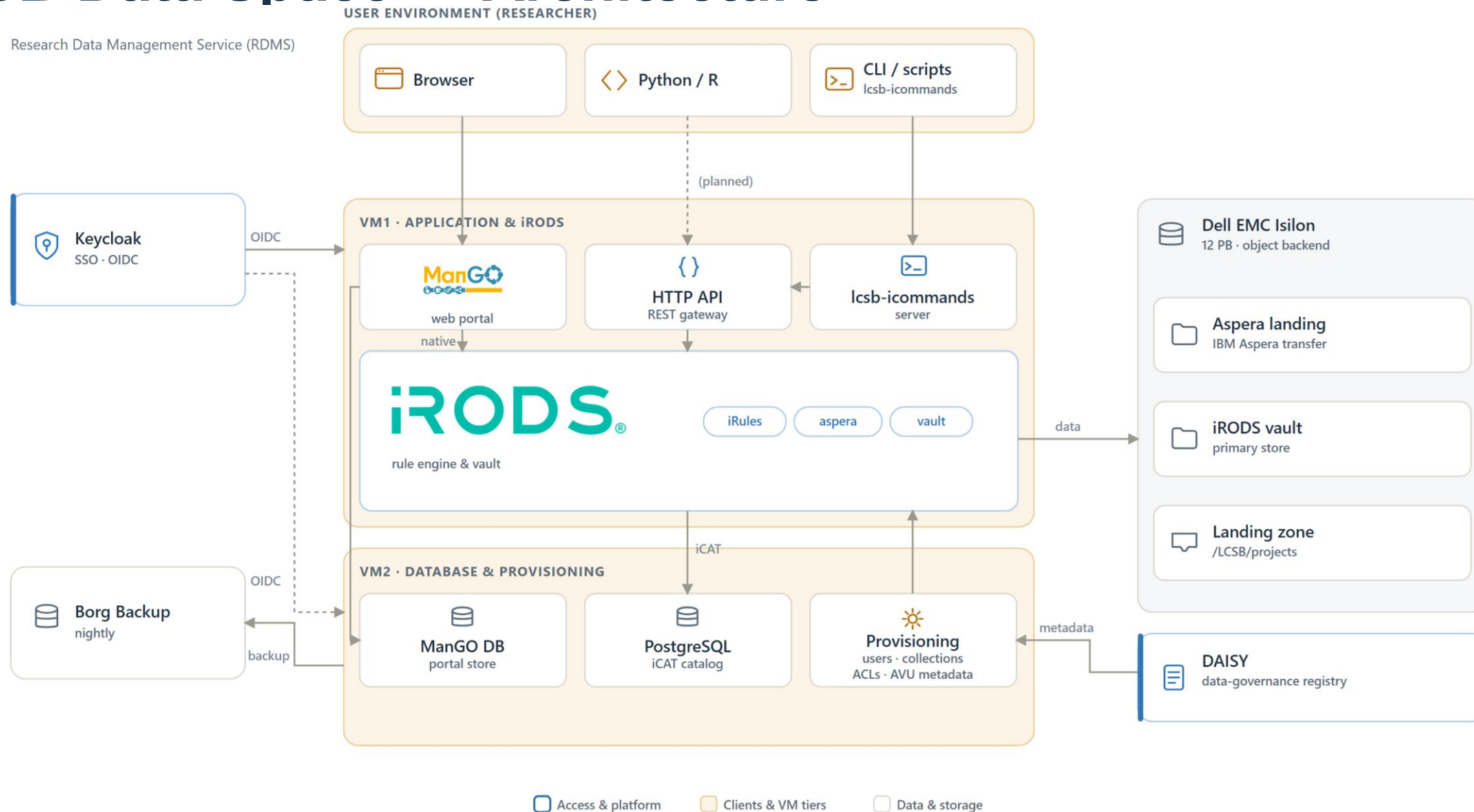


Rich AVU Metadata

Attribute-value metadata drives discovery, access control, and lifecycle.

BUILT ON
iRODS®

LCSB Data Space — Architecture



Authentication & Provisioning

1

Sign in with Keycloak (OIDC)

SSO for all services · Device Flow (RFC 8628) for CLI & HPC

2

Brokered via the iRODS HTTP API

No raw iRODS credentials are ever exposed

3

Project membership from DAISY

DAISY is the source of truth for access

4

Auto-provisioned on first login

iRODS user, project collections & ACLs created automatically

5

Synced on every login

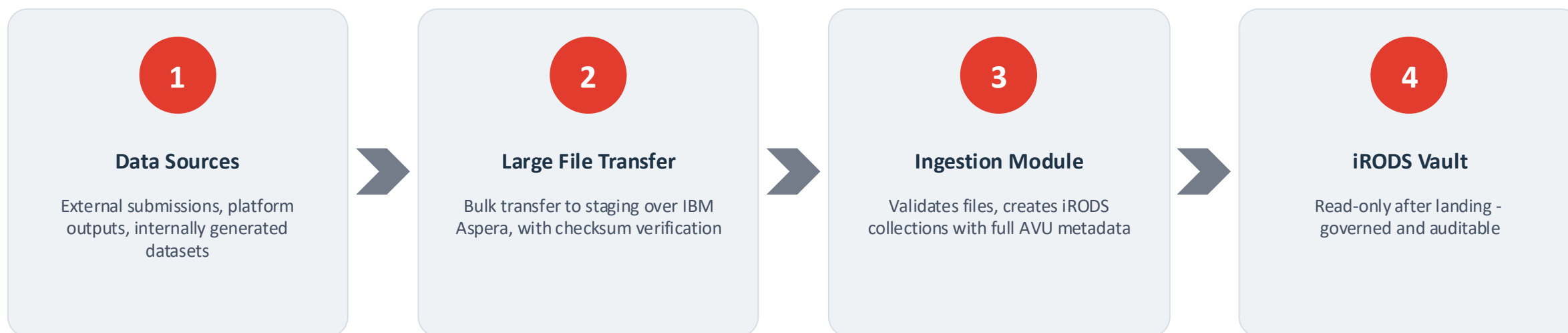
Project membership re-synced from DAISY in the background

```
iRODS collection structure

/LCSB/
├─ home/{user}/
├─ landing/{user}/{project}/
└─ projects/{ACRONYM}/
    ├─ data/raw/
    ├─ data/processed/
    └─ {dataset_id}/data/...
```

Created on first login — defines the scope for the project's data.

Ingestion Pipeline

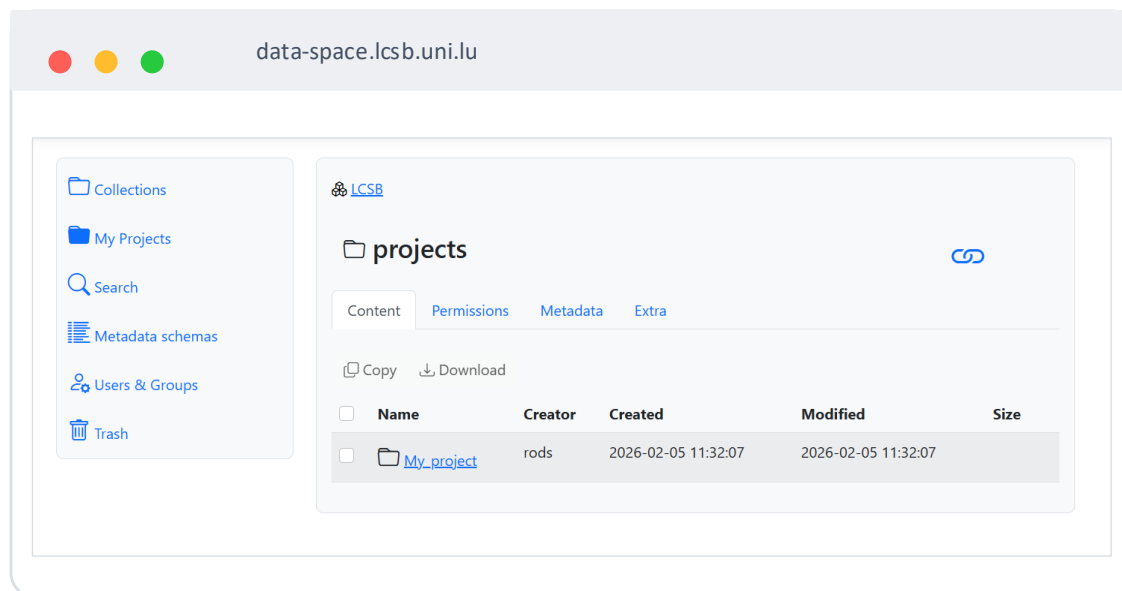


Multiple ingestion sources supported — no manual steps required from the researcher

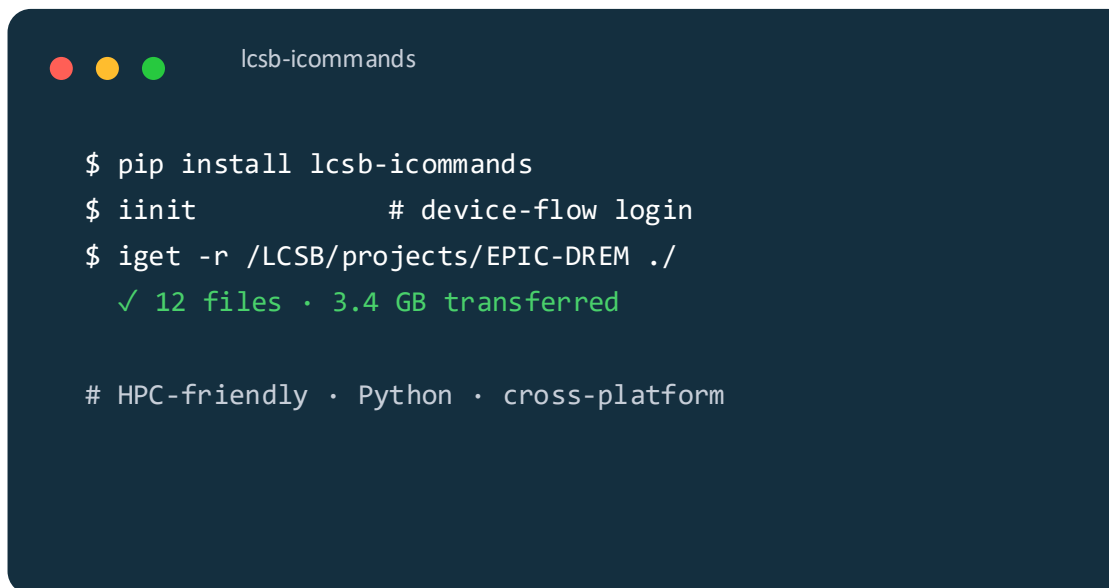
Accessing Your Data



portal · by KU Leuven



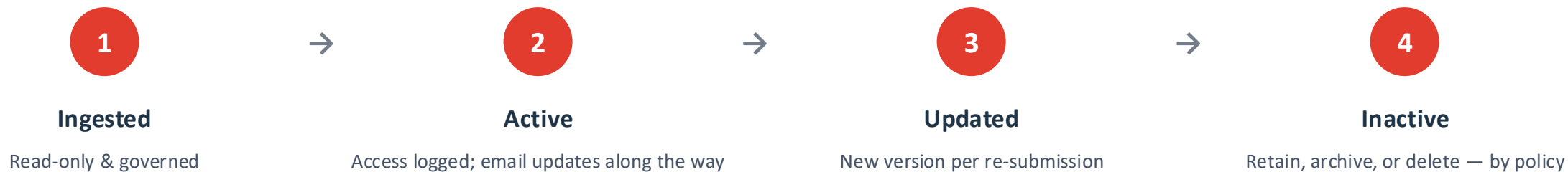
lcsb-icommands · our CLI, built on iRODS icommands — command line



Programmatic access via Python/R – coming soon

All use university Keycloak SSO — no separate account needed

Dataset Lifecycle



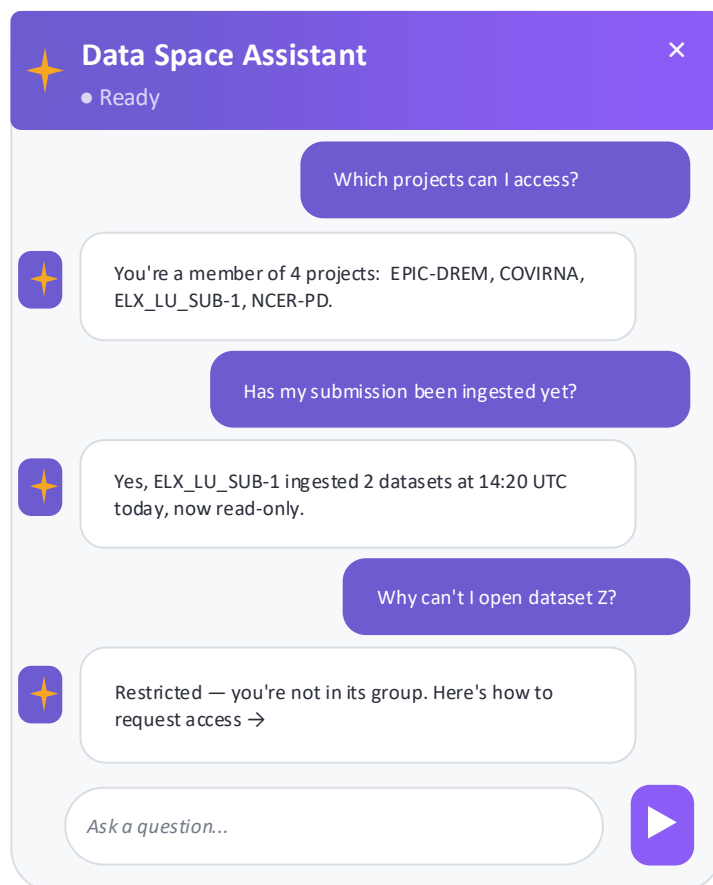
Live Today

- Email notification on ingestion — researcher informed when data lands
- Every access logged, full audit trail
- Retention follows the legal basis

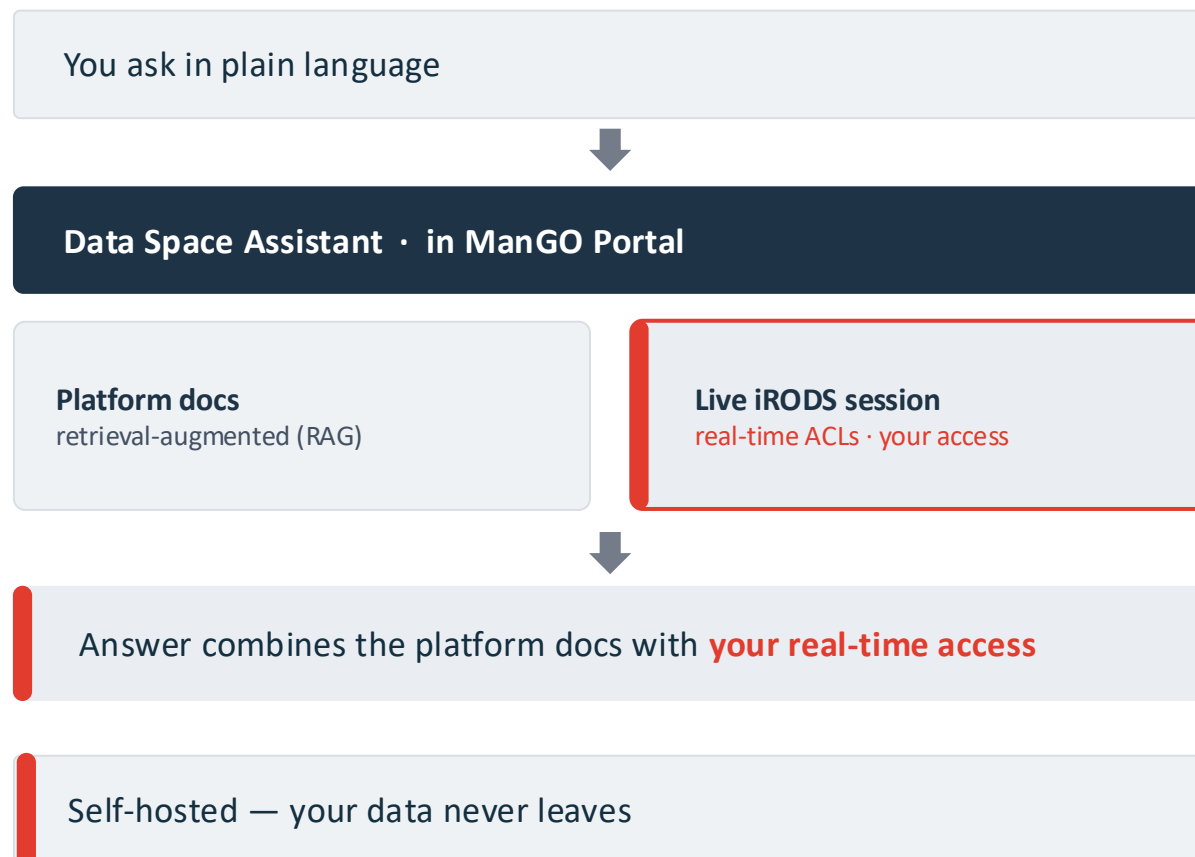
In Development

- Automatic alerts as a dataset nears its retention/expiry date
- Dataset versioning — a re-submission creates a new version
- Deletion workflow — steward-approved, leaving an auditable deprecation record

AI Chatbot Assistant



How it works



Lessons Learned

Technical

- 1 iRODS 5 migration — plan for breaking changes in python-irodsclient and the HTTP API
- 2 OIDC token lifecycle is the hard part — plan refresh and expiry across every service

Operational

- 1 Get cross-team agreement early — it saves a lot of rework
- 2 External systems are slow — sync in the background, never block the user

Live Demo

What We Will Show

- Browse projects and datasets in the web portal
- View the metadata attached to a dataset
- Access the same data from the command line
- Ask the AI assistant, grounded in your real access

data-space.lcsb.uni.lu



LCSB Research Infrastructure

LCSB Data Space

A secure, structured platform for managing your research data throughout its full lifecycle — from collection to publication.



Secure & Compliant

GDPR-aligned storage with full audit trail and access control



Project-Based Organisation

Data organised by DAISY projects with rich metadata



Full Data Lifecycle

From upload and review to vaulting and archival



AI-Powered Assistant

Ask questions about your data in natural language

Welcome back

Sign in with your LCSB university account to access your research data.

[Login with LCSB SSO](#)

[New to Data Space? Read the documentation](#)

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Questions?

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